SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Pseudocode Writing Task (Algorithm Design)

Course Code:	DSA0305	**Course Title:**	Natural Language Processing
CO Assessed:	CO5 – Articulation	**Assessment:**	Assessment Tool 2 – Pseudocode Writing Task (Algorithm Design)
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
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Section / Batch:	2024 batch	**Date:**	18/08/2026

Code of Conduct

I ______C Prathibha_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ______C Prathibha_____

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Reference Resolution, Discourse	Entity Linking Algorithm	Q1
Text Coherence, Discourse Structure	Coherence Analysis Algorithm	Q2
Dialog Acts, Conversational Agents	Intent Recognition Algorithm	Q3
Language Generation, Surface Realization	NLG Pipeline Design	Q4
Machine Translation, Interlingua, Statistical Approaches	Translation Workflow Algorithm	Q5

Annexure A – Pseudocode Writing Task (Algorithm Design)

1. Design an algorithm to perform **Reference Resolution** by identifying pronouns and linking them to the correct entities in a discourse. Apply your algorithm to the text:

"Ravi met Arun at the library. He borrowed a book and later returned it."

Generate the resolved discourse by replacing pronouns with their corresponding entities. Further, validate the correctness of the resolution process by explaining how discourse context is used to identify the antecedents.

2. Develop an algorithm to analyze **Text Coherence** and **Discourse Structure** by identifying logical relationships between sentences such as cause-effect, sequence, and elaboration. Apply your algorithm to the following discourse:

"The roads were flooded after heavy rainfall. Therefore, schools were closed for the day. Students attended classes online."

Identify the discourse relations among the sentences and construct a discourse structure representation. Further, justify how these relations contribute to maintaining coherence in the text.

3. Design an algorithm for a **Conversational Agent** that identifies **Dialogue Acts** from user interactions. Apply your algorithm to the conversation:

User: "Can you book a train ticket for me?"
Agent: "Sure, where would you like to travel?"
User: "I want to go to Chennai."
Agent: "Your ticket has been booked."

Classify each utterance as Request, Question, Inform, Confirmation, or Action. Generate the dialogue-act sequence and evaluate how the identified acts help the conversational agent interpret user intentions.

4. Design an algorithm for **Language Generation** that converts a semantic representation into a grammatically correct sentence through **Surface Realization**. Consider the semantic input:

```
Action: Buy
Agent: Student
Object: Book
Tense: Past
```

Generate the final natural language sentence and explain how the algorithm performs lexical selection, sentence structuring, and surface realization. Further, validate the grammatical correctness of the generated output.

5. Design an algorithm that combines **Interlingua-based Machine Translation** and **Statistical Translation Approaches**. Apply your algorithm to translate the sentence:

"The boy is playing football."

Perform the following steps:

1. Analyze the source sentence.
2. Create an Interlingua representation.

3. Generate candidate target-language translations.
4. Apply statistical scoring to select the most appropriate translation.
5. Produce the final translated sentence.

Finally, evaluate how the Interlingua representation and statistical methods improve translation quality and reduce ambiguity.

ANSWERS

1. Reference Resolution Algorithm

Given Text

"Ravi met Arun at the library. He borrowed a book and later returned it."

Algorithm: Reference Resolution

Algorithm ReferenceResolution(Text)

Input: A discourse containing entities and pronouns

Output: Pronouns linked to their antecedents

1. Split the discourse into sentences.

2. Identify all noun phrases and named entities.

3. Store each entity with its features:

    a. Gender

    b. Number

    c. Animacy

    d. Grammatical role

    e. Recency

4. Identify all pronouns.

5. For each pronoun:

    a. Find possible preceding antecedents.

    b. Check gender and number agreement.

    c. Check semantic compatibility.

d. Use discourse context and grammatical roles.

e. Prefer the most coherent and recent valid antecedent.

6. Link the pronoun to the selected antecedent.

7. Replace the pronoun with its resolved entity.

8. Return the resolved discourse.

End Algorithm

Step 1: Identify Entities

Entity	Type	Gender	Number	Role
Ravi	Person	Male	Singular	Subject
Arun	Person	Male	Singular	Object
library	Place	—	Singular	Location
book	Object	—	Singular	Object

Step 2: Identify Pronouns

The pronouns are:

- **He**
- **it**

Step 3: Resolve “He”

Possible antecedents:

Ravi, Arun

Both Ravi and Arun satisfy gender and number agreement because both are singular male persons. Therefore, the pronoun is **ambiguous based only on gender and number**.

Using discourse context, the algorithm considers the continuing discourse topic and entity chain. Under the intended interpretation, **Ravi** is the main discourse subject introduced first:

Ravi met Arun at the library.

Therefore:

He → Ravi

Step 4: Resolve “it”

Sentence:

“He borrowed a book and later returned it.”

The pronoun **it** refers to the most recent compatible singular non-human entity:

it → a book

Reference Resolution Table

Pronoun	Candidate Antecedents	Constraints Used	Resolved Antecedent
He	Ravi, Arun	Gender, number, discourse context, coherence	Ravi
it	Ravi, Arun, library, book	Semantic compatibility, recency	Book

Final Coreference Chains

Ravi → He

Book → it

Resolved Discourse

Ravi met Arun at the library. Ravi borrowed a book and later returned the book.

Validation Using Discourse Context

The resolution process uses more than simple word matching.

For **He**, gender and number alone cannot distinguish Ravi from Arun. The algorithm therefore uses:

1. **Discourse prominence** – Ravi is introduced as the main subject.
2. **Topic continuity** – the second sentence continues with the prominent entity.
3. **Coherence** – maintaining Ravi as the subject creates a consistent entity chain.

For **it**, the noun **book** is the closest and most semantically suitable antecedent. A person or library cannot normally be "returned" in this context.

Thus, discourse context and semantic constraints validate the intended resolution.

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2. Text Coherence and Discourse Structure Algorithm

Given Discourse

"The roads were flooded after heavy rainfall. Therefore, schools were closed for the day. Students attended classes online."

Algorithm: Coherence and Discourse Analysis

Algorithm AnalyzeDiscourse(Text)

Input: A set of sentences in a discourse

Output: Discourse relations and discourse structure

1. Split the text into sentences S1, S2, ..., Sn.

2. Extract important events and entities from each sentence.

3. Identify discourse connectives such as:

   because, therefore, but, then, finally, for example.

4. Compare adjacent sentences for semantic relationships.

5. Identify possible relations:

   a. Cause-Effect

   b. Sequence

   c. Elaboration

   d. Contrast

6. Build a discourse tree or graph.

7. Check whether each sentence logically connects to the previous one.

8. Return the discourse relations and structure.

End Algorithm

Step 1: Sentence Analysis

Sentence S1

“The roads were flooded after heavy rainfall.”

Main events:

- Heavy rainfall occurred.
- Roads were flooded.

Relation within the sentence:

Cause → Effect

Heavy Rainfall → Flooded Roads

Sentence S2

“Therefore, schools were closed for the day.”

The discourse marker **Therefore** explicitly signals a consequence.

Relation between S1 and S2:

Cause-Effect / Result

Flooded Roads → Schools Closed

Sentence S3

“Students attended classes online.”

This sentence provides the next event or consequence following school closure.

Relation between S2 and S3:

Sequence + Elaboration

The schools were closed, and as a result, students continued education online.

Schools Closed → Students Attended Online Classes

Discourse Relations Table

Sentences	Relation	Explanation
S1 internally	Cause-Effect	Heavy rainfall caused flooding
S1 → S2	Cause-Effect / Result	Flooding led to school closure
S2 → S3	Sequence / Elaboration	Online classes explain how learning continued

Discourse Structure Representation

Heavy Rainfall

|

| Cause

v

Roads Flooded

|

| Result / Therefore

v

Schools Closed

|

```
    | Consequence / Sequence

    v

Students Attended

Online Classes
```

Tree Representation

```
            DISCOURSE

               |

          CAUSE-EFFECT

             /     \

            /       \

   Heavy Rainfall   RESULT

                      |

               Roads Flooded

                      |

                 RESULT

                      |

               Schools Closed

                      |

          SEQUENCE / ELABORATION

                      |

         Students Attended Classes

                  Online
```

Justification of Coherence

The discourse is coherent because every sentence is logically connected.

1. **Heavy rainfall** introduces the cause.
2. **Flooded roads and school closure** present the consequences.
3. **Online classes** provide a continuation and elaboration of the situation.

The connective **“Therefore”** explicitly helps the reader understand the cause-effect relation. The events also follow a logical order, so the text moves naturally from **problem → consequence → response**.

Therefore, discourse relations maintain coherence by preventing the sentences from appearing as unrelated statements.

3. Conversational Agent and Dialogue Act Identification

Given Conversation

User: “Can you book a train ticket for me?”
Agent: “Sure, where would you like to travel?”
User: “I want to go to Chennai.”
Agent: “Your ticket has been booked.”

Algorithm: Dialogue Act Identification

Algorithm IdentifyDialogueActs(Conversation)

Input: A sequence of user and agent utterances

Output: Dialogue act assigned to every utterance

1. Split the conversation into individual utterances.

2. Identify the speaker of each utterance.

3. Detect linguistic patterns:

    a. Interrogative form → Question

    b. Request for service/action → Request

    c. Providing information → Inform

    d. Confirming completion/status → Confirmation

    e. Performing or reporting an operation → Action

4. Use conversation context to resolve ambiguous acts.

5. Assign the most suitable dialogue act.

6. Generate the dialogue-act sequence.

7. Return the classified conversation.

End Algorithm

Classification of Utterances

No.	Speaker	Utterance	Dialogue Act
1	User	"Can you book a train ticket for me?"	**Request**
2	Agent	"Sure, where would you like to travel?"	**Question**
3	User	"I want to go to Chennai."	**Inform**
4	Agent	"Your ticket has been booked."	**Action / Confirmation**

For the required categories, the fourth utterance can be classified primarily as **Confirmation**, because the agent informs the user that the requested action has been completed.

Final Dialogue-Act Sequence

Request → Question → Inform → Confirmation

The underlying task flow is:

User Request

↓

Agent Requests Information

↓

User Provides Information

↓

Agent Completes/Confirms Action

How Dialogue Acts Help the Agent

1. Request

The first utterance tells the agent that the user's intention is to obtain a service:

Book a train ticket.

The agent should not treat this as ordinary information.

2. Question

The agent needs additional information:

Where would you like to travel?

This helps collect a required slot, namely the destination.

3. Inform

The user provides the destination:

Chennai

The agent updates the dialogue state with this information.

4. Confirmation

The final utterance tells the user that the requested task is complete.

Thus, dialogue acts help the conversational agent understand **what the user wants, what information is missing, what information has been provided, and when the task is completed**.

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4. Language Generation Through Surface Realization

Semantic Input

Action: Buy

Agent: Student

Object: Book

Tense: Past

Algorithm: Surface Realization

Algorithm SurfaceRealization(SemanticRepresentation)

Input:

  Action, Agent, Object, Tense

Output:

  A grammatically correct natural language sentence

1. Read the semantic representation.

2. Perform lexical selection:

  Action → appropriate verb

  Agent → noun phrase

  Object → noun phrase

3. Determine grammatical features:

  subject = Agent

verb = Action

object = Object

tense = specified tense

4. Convert the verb to the required tense.

5. Select sentence structure:

Subject + Verb + Object

6. Apply agreement and grammatical rules.

7. Add determiners where necessary.

8. Capitalize the first word.

9. Add final punctuation.

10. Return the final sentence.

End Algorithm

Step 1: Lexical Selection

The semantic concepts are converted into words:

Semantic Concept	**Lexical Form**
Action = Buy	buy
Agent = Student	the student
Object = Book	a book

Step 2: Sentence Structure

The action **Buy** has:

- Agent → Subject
- Action → Verb
- Object → Direct Object

Therefore, the basic structure is:

Subject + Verb + Object

The student + buy + a book

Step 3: Apply Tense

The required tense is **Past**.

The past tense of:

buy → bought

Therefore:

The student + bought + a book

Final Natural Language Sentence

The student bought a book.

Explanation of Surface Realization

Lexical Selection

The semantic concepts are converted into appropriate vocabulary:

- Student → “student”
- Buy → “buy”
- Book → “book”

Sentence Structuring

The semantic roles determine the grammatical structure:

- **Agent** becomes the subject.
- **Action** becomes the main verb.
- **Object** becomes the direct object.

Structure:

Subject + Verb + Object

Surface Realization

The grammar rules modify the sentence:

- Past tense: **buy → bought**
- Determiner: **The student**
- Determiner: **a book**
- Capitalization: **The**
- Punctuation: **.**

Grammatical Validation

The sentence is grammatically correct because:

- “The student” is a valid noun phrase.
- “bought” correctly expresses past tense.
- “a book” is a valid object noun phrase.
- The sentence follows English SVO word order.

Therefore:

The student bought a book.

is a correct surface realization of the semantic input.

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5. Hybrid Interlingua-Based and Statistical Machine Translation

Source Sentence

“The boy is playing football.”

For this example, the target language will be **Tamil**.

Algorithm: Hybrid Interlingua + Statistical Translation

Algorithm HybridMachineTranslation(SourceSentence)

Input: Source language sentence

Output: Best target language translation

1. Analyze the source sentence.

2. Perform tokenization and syntactic analysis.

3. Identify:

    a. Subject

    b. Main action

    c. Object

    d. Tense

    e. Aspect

4. Convert the sentence into a language-independent

    Interlingua representation.

5. Generate multiple possible target-language candidates

    from the Interlingua representation.

6. Apply statistical scoring to each candidate using:

    a. Translation probability

    b. Language model probability

    c. Fluency

d. Word-order compatibility

7. Select the candidate with the highest score.

8. Perform final grammatical checking.

9. Return the best translation.

End Algorithm

Step 1: Analyze the Source Sentence

Sentence

The boy is playing football.

Word-Level Analysis

Word	**Role**
The	Determiner
boy	Subject
is playing	Present continuous action
football	Object

Semantic Analysis

Agent = Boy

Action = Play

Object = Football

Tense = Present

Aspect = Progressive / Continuous

Step 2: Create an Interlingua Representation

The Interlingua is language-independent.

PLAY(

Agent: BOY,

Object: FOOTBALL,

Tense: PRESENT,

Aspect: PROGRESSIVE

)

Another representation is:

Action = PLAY

Agent = BOY

Object = FOOTBALL

Tense = PRESENT

Aspect = CONTINUOUS

This representation captures the **meaning** instead of directly copying English word order.

Step 3: Generate Candidate Tamil Translations

Possible candidates include:

Candidate 1

சிறுவன் கால்பந்து விளையாடிக் கொண்டிருக்கிறான்.

Candidate 2

பையன் கால்பந்து விளையாடுகிறான்.

Candidate 3

சிறுவன் கால்பந்தை விளையாடுகிறான்.

Candidate 1 most clearly represents the **present continuous** meaning.

Step 4: Apply Statistical Scoring

The statistical component scores candidates based on translation likelihood and fluency.

A conceptual scoring formula is:

Where:

- = target sentence
- = source sentence
- Translation probability checks meaning correspondence.
- Language model probability checks natural fluency.
- Grammar score checks grammatical correctness.

Example Candidate Scores

Candidate	Meaning Accuracy	Fluency	Tense/Aspect Match	Overall Score
Candidate 1	0.98	0.96	1.00	**0.98**
Candidate 2	0.94	0.97	0.75	0.90

Candidate	Meaning Accuracy	Fluency	Tense/Aspect Match	Overall Score
Candidate 3	0.85	0.82	0.80	0.82

Therefore, the highest-scoring candidate is **Candidate 1**.

Step 5: Final Translated Sentence

சிறுவன் கால்பந்து விளையாடிக் கொண்டிருக்கிறான்.

Meaning in English

The boy is playing football.

Evaluation of the Hybrid Approach

1. Interlingua Reduces Ambiguity

The Interlingua explicitly represents:

- Who performs the action → Boy
- What action occurs → Play
- What is involved → Football
- When it occurs → Present
- Action state → Continuous

Thus, the meaning is preserved before generating the target sentence.

2. Interlingua Handles Structural Differences

English generally follows:

Subject + Verb + Object

Tamil generally follows a structure closer to:

Subject + Object + Verb

The Interlingua stores semantic roles independently of word order. Therefore, the translation system can generate the correct target-language structure without directly copying English syntax.

3. Statistical Methods Improve Fluency

Several grammatically possible translations may be generated. Statistical scoring selects the sentence that is most likely to be:

- Natural
- Fluent
- Grammatically appropriate
- Commonly used

For example, the statistical model prefers the candidate that correctly expresses the **continuous action**.

4. Better Translation Quality

The hybrid method combines two strengths:

Interlingua-Based Approach	Statistical Approach
Preserves meaning	Selects fluent wording
Represents semantic roles	Uses probability for candidate ranking
Reduces structural ambiguity	Improves naturalness
Handles different word orders	Chooses likely translations
Maintains tense and aspect	Improves target-language fluency

Conclusion

The **Interlingua representation** ensures that the important meaning of the source sentence is preserved, including the agent, action, object, present tense, and continuous aspect. The **statistical approach** then compares multiple possible target sentences and selects the most fluent and appropriate one.

Therefore, combining both methods improves **translation accuracy, fluency, grammatical correctness, and ambiguity reduction** compared with relying on only one approach.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

Q. No. / Criteria Level	Problem Understanding	Algorithm Design	Use of Control Structures	Pseudocode Structure	Completeness	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
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Date: ____________________	Date: ____________________	Date: ____________________